Detection & AI Methodology

CABINGUARD-ADI: A Zero-Compliance Multimodal In-Cabin Framework for Prodromal Prediction and Fail-Safe Actuation in Acute Driver Incapacitation

TSYP14 Technical Challenge — Multimodal Medical-Event Detection and Deterministic Safety Reasoning

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1. Detection Objective

CABINGUARD-ADI is designed to distinguish four operating classes:

1. **Class 0: Nominal Driving**
2. **Class 1: Arrhythmic Syncope**
3. **Class 2: Generalized Tonic-Clonic Seizure**
4. **Class 3: Cognitive-Autonomic Deficit**

The system is not intended to diagnose chronic disease. Its objective is to detect acute multimodal patterns that may indicate loss of safe vehicle control.

2. Dataset Roles and Evidence Boundaries

The available public datasets do not provide a synchronized clinical cohort containing the three target acute medical events inside a moving vehicle. They are therefore assigned only to the modalities and behaviors they genuinely support.

Dataset	Verified / Intended Modality	Role in CABINGUARD-ADI
In_Cabin_Behavior_Driver (Hugging Face)	Visible, gated access repository	In-cabin seated-driver visual baseline. Used for cabin geometry, head/body behavior and domain-specific video analysis after access. No medical-event label is assumed because the public dataset card does not expose the label structure.
Ala Mahlak — Driver Monitoring Dataset (Kaggle)	Driver-monitoring visual data	Auxiliary visual baseline for driver-monitoring features. Exact classes must be verified from the downloaded files before any class-specific claim is made.
Advance Driver Monitoring System (ADMS) Dataset (Kaggle)	Labeled driver-behavior images	Nominal/distractor modeling using the documented behaviors: eye closure, yawning, seat-belt use, smoking and mobile-phone use. It is not treated as a medical-event dataset.
Radar-Based Vital Sign Monitoring (Kaggle)	FMCW radar cardiorespiratory data	Auxiliary non-contact vital-sign processing reference. The linked dataset concerns baby vital-sign monitoring and is therefore not treated as an in-cabin driver or acute-medical-event dataset.
ADL-Dataset (Kaggle)	Pending direct inspection of downloaded contents	Retained only as a candidate activity/posture source. No modality or label claim is made until its files are inspected.

The dataset strategy therefore separates:

$$\boxed{\text{Real baseline observations}} \quad + \quad \boxed{\text{Explicit synthetic feature-level event injection}}$$

rather than falsely labeling normal driver-monitoring samples as clinical syncope, seizure, or neuroglycopenic events.

3. Multimodal Input Features

The detector uses only features supported by the proposed sensor architecture:

- **Optical / NIR:** head pitch, head roll, gaze, PERCLOS, wrist motion, pallor/rPPG cues and NIR diaphoresis cues.
- **Tactile seat:** Center of Pressure, lumbar contact and quadrant dynamics.
- **Mechanical cardiac / respiratory:** BCG-derived inter-beat intervals, respiratory rate and signal quality.
- **Vehicle interaction:** EPS steering torque and steering-torque rate.

Visual datasets contribute only to the optical and behavioral feature space. Radar/vital-sign data contribute only to cardiorespiratory processing studies. Missing modalities are not silently fabricated from unrelated datasets.

4. Baseline Normalization and Personalized Calibration

For driver/session i and signal s, the robust baseline center and MAD scale are:

$$c_{i,s} = \text{median}(x_{i,s,t}), \qquad \sigma_{i,s} = 1.4826 \text{ median} \left| x_{i,s,t} - c_{i,s} \right|. \tag{1}$$

The personalized center is regularized toward the population prior:

$$\tilde{c}_{i,s} = w_i c_{i,s} + (1 - w_i)\mu_{\text{pop},s}, \qquad w_i = \frac{N_{\text{good}}}{N_{\text{good}} + \kappa}, \tag{2}$$

with $\kappa = 300$. The session baseline is frozen at $t = 60$ s.

The orientation-aware deviation used by the detector is:

$$z_{i,s,t} = O_s \frac{x_{i,s,t} - \tilde{c}_{i,s}}{\tilde{\sigma}_{i,s}}, \tag{3}$$

where $O_s \in \{-1, +1, \text{bidir}\}$ aligns deterioration toward positive risk.

5. Feature-Level Scenario Construction

The public datasets provide real baseline behavior, while the target acute medical events are represented by controlled transformations of the **derived feature trajectories**. The transformations below are proposed simulation operators, not clinical measurements.

Let $x_s^{(b)}(t)$ denote a real baseline feature extracted from a dataset. A synthetic scenario feature is defined as:

$$x_s^{(\mathrm{syn})}(t) = x_s^{(b)}(t) + \alpha_s \, g_k(t), \tag{4}$$

where:

- α_s is the configured scenario magnitude for feature s;
- $g_k(t)$ is the temporal envelope for scenario class k;
- the original baseline $x_s^{(b)}(t)$ remains unchanged and traceable.

A generic prodrome-to-acute envelope can be written as:

$$g_k(t) = \begin{cases} 0, & t < t_{p,k}, \\ \dfrac{t - t_{p,k}}{t_{a,k} - t_{p,k}}, & t_{p,k} \le t < t_{a,k}, \\ 1, & t \ge t_{a,k}, \end{cases} \tag{5}$$

where $t_{p,k}$ is the configured prodromal onset and $t_{a,k}$ is the configured acute-event onset.

For features that should move toward a specific acute target value $x_{s,k}^*$, interpolation is used instead of unconstrained addition:

$$x_s^{(\mathrm{syn})}(t) = \big(1 - g_k(t)\big) x_s^{(b)}(t) + g_k(t) x_{s,k}^*. \tag{6}$$

This operator is appropriate for features such as head pitch, lumbar contact, heart rate, or steering torque when the scenario specification defines a target range or endpoint.

5.1 Oscillatory Motor Features

For seizure-related clonic motion or cognitive-autonomic tremor, an oscillatory component can be injected at the feature level:

$$x_s^{(\mathrm{syn})}(t) = x_s^{(b)}(t) + A_s \, g_k(t) \sin\left(2\pi f_k t + \phi\right), \tag{7}$$

where f_k is selected only from the scenario ranges already defined by the CABINGUARD-ADI methodology:

$$f_{\mathrm{seizure}} \in [3, 6] \text{ Hz}, \qquad f_{\mathrm{tremor}} \in [4, 8] \text{ Hz}.$$

No new clinical frequency range is introduced by this transformation.

5.2 Signal-Quality Degradation

To simulate sensor degradation or temporary loss without altering the original source sample:

$$x_s^{(\mathrm{deg})}(t) = m_s(t) \, x_s^{(b)}(t) + \big(1 - m_s(t)\big) \eta_s(t), \tag{8}$$

where $m_s(t) \in \{0, 1\}$ is an availability mask and $\eta_s(t)$ represents the configured degraded/noise condition used by the test scenario.

The corresponding source-availability score is:

$$A_{\text{sources},t} = \frac{1}{N_s} \sum_{s=1}^{N_s} m_s(t). \tag{9}$$

This allows missing-data and fail-silent behavior to be tested explicitly rather than silently imputing nonexistent medical evidence.

6. Scenario Labels for Model Development

Each synthetic trajectory receives a provenance-preserving phase label:

$$p_k(t) = \begin{cases} \texttt{NOMINAL}, & t < t_{p,k}, \\ \texttt{PRODROMAL}, & t_{p,k} \le t < t_{a,k}, \\ \texttt{ACUTE}, & t \ge t_{a,k}. \end{cases} \tag{10}$$

The etiology label remains:

$$y_k \in \{\texttt{SYNCOPE}, \texttt{SEIZURE}, \texttt{COGNITIVE_AUTONOMIC}, \texttt{DISTRACTOR}\}.$$

Every generated training/evaluation sample should therefore retain:

$$\mathcal{R} = \{\text{source dataset}, \text{source sample}, \text{real/synthetic flag}, y_k, p_k(t), \text{injected features}\},$$

so real observations and synthetic modifications remain distinguishable throughout evaluation.

7. Multi-Etiology Detection Signatures

Class	**Prodromal / Acute Pattern**	**Optical & Seat Features**	**Cardiac / EPS Features**
Nominal	Continuous baseline	Active saccades; $\theta_{\text{pitch}} \in [-10°, 15°]$; PERCLOS < 0.15; stable CoP; $R_{\text{lumbar}} > 85\%$.	Steady personalized HR baseline and active steering micro-corrections.
Arrhythmic Syncope	Presyncope followed by flaccid collapse	Pallor shift; gaze narrowing; acute $\theta_{\text{pitch}} > 45°$; lumbar detachment; anterior CoP shift.	Tachypnea/IBI volatility; acute bradycardia/asystole pattern; loss of voluntary steering torque.
Tonic-Clonic Seizure	Focal arrest followed by tonic/clonic activity	Kinematic freeze; unblinking stare; 3–6 Hz head/wrist oscillation; extreme quadrant dynamics.	Autonomic tachycardia; severe BCG degradation during convulsion; erratic high steering torque.
Cognitive-Autonomic Deficit	Progressive autonomic/cognitive deterioration	NIR diaphoresis; 4–8 Hz microtremor; gaze fixation; upright posture may persist.	Tachycardia relative to baseline; loss of voluntary steering micro-corrections.

8. Causal Temporal Detection

A causal $W = 3.0$ s window classifies each feature trajectory as:

- `TRANSIENT`;
- `PERSISTENT`, when more than 80% of the window remains above `ELEVATED`;

- `WORSENING`, when the risk trajectory has $\dot{z}_{i,s,t} > 0$.

Short distractor behavior should therefore be evaluated against the persistence requirement rather than labeled as medical deterioration by construction.

9. Cross-Modal Plausibility and Uncertainty

If one modality physically contradicts another, the engine sets:

$$P_{\text{conflict},t} = 1.$$

Total uncertainty is:

$$U_t = 1 - \left(w_f C_{\text{fusion},t} + w_a A_{\text{sources},t}(1 - P_{\text{conflict},t})\right), \tag{11}$$

with:

$$w_f = 0.65, \qquad w_a = 0.35.$$

Contradictory or insufficient evidence therefore supports abstention rather than forced classification.

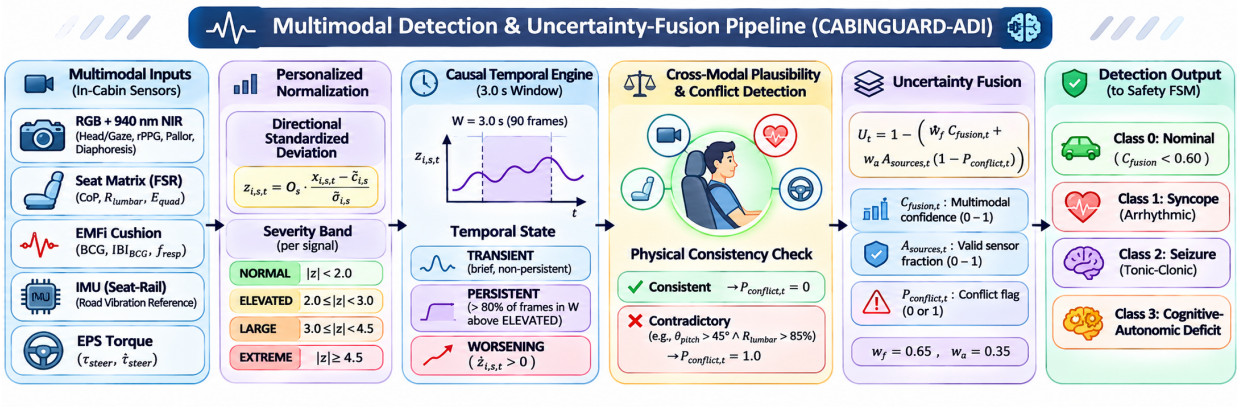


Figure 1: Original CABINGUARD-ADI multimodal detection and uncertainty-fusion infographic retained unchanged.

10. Detection-to-Decision Logic

The deterministic decision mechanism uses:

- **Level 1:** nominal monitoring;
- **Level 2:** prodromal verification using the 4.0 s Cognitive Handshake;
- **Level 3:** emergency MRM for high-confidence persistent/worsening evidence;
- **Fail-Silent:** abstention when conflict, insufficient source availability, or high uncertainty prevents a defensible decision.

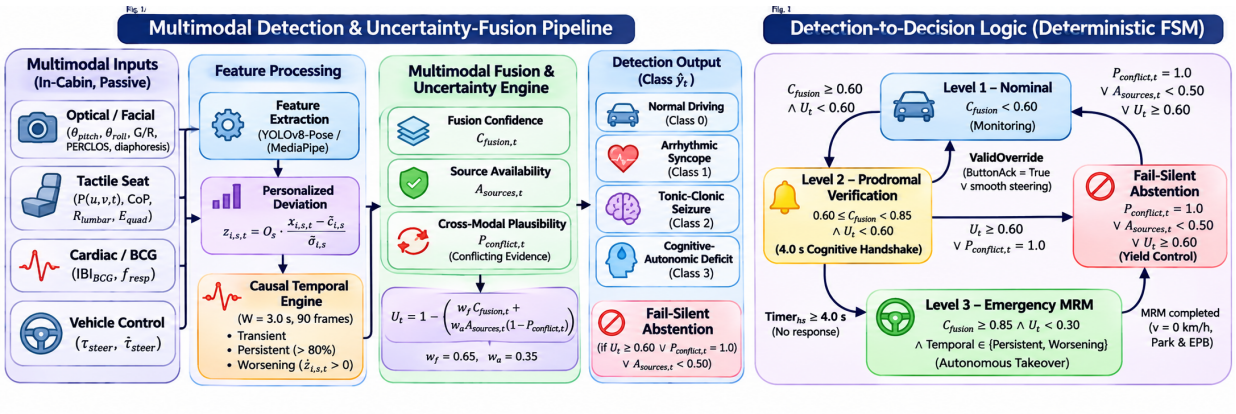


Figure 2: Original CABINGUARD-ADI detection-to-decision and deterministic FSM infographic retained unchanged.

11. Training / Evaluation Strategy

The model-development data should be partitioned by original subject/session whenever subject identifiers are available. Synthetic derivatives originating from one baseline sample must remain in the same partition as that source sample to avoid leakage between training and evaluation.

The data used by the classifier can therefore be represented as:

$$\mathbf{X}_t = [z_{\text{head}}, z_{\text{gaze}}, z_{\text{PERCLOS}}, z_{\text{CoP}}, z_{\text{lumbar}}, z_{\text{HR}}, z_{\text{resp}}, z_{\text{IBI}}, z_{\tau}, A_{\text{sources}}, P_{\text{conflict}}]_t. \tag{12}$$

The exact learning model and hyperparameters must be reported only after an implemented training pipeline is selected. This document does not invent an untested architecture.

12. Planned Performance Metrics

The following quantities remain **to be measured**:

- class-wise precision, recall and F1-score;
- prodromal lead time;
- acute detection latency;
- false-positive / unintended MRM rate;
- false-negative rate;
- distractor rejection;
- conflict-abstention rate;
- runtime latency.

No numerical performance value is claimed before execution of the reproducible evaluation.

13. False-Positive and False-Negative Considerations

False positives are addressed through personalized calibration, multimodal agreement, temporal persistence, distractor examples, Cognitive Handshake verification and conflict-based abstention.

False negatives are addressed through complementary optical, postural, cardiac, respiratory and steering evidence so that degradation of one modality does not automatically remove all evidence.

Dataset-specific limitations remain explicit: driver-monitoring datasets primarily model normal, drowsy or distracted behavior, while the target acute medical events remain synthetic feature-level scenarios until clinically representative in-cabin data become available.

14. Validation Status

The complete quantitative benchmark has not yet been executed. The evaluation requires repeated multimodal feature extraction, synthetic event generation, temporal fusion, fault injection and cross-dataset assessment. Full testing is currently pending access to higher-compute data-center resources.

Until then, the equations above define the **proposed scenario-construction methodology**; they are not presented as experimentally validated transformations or clinical physiological models.